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Analysis of Software Developers' Programming Language Preferences and Community Behavior From Big5 Personality Traits

Md. Saddam Hossain Mukta[1] | Badrun Nessa Antu[2] | Nasreen Azad[1] | Iftekharul Abedeen[2] | Najmul Islam[1]

[1]LUT School of Engineering Sciences, Lappeenranta-Lahti University of Technology, Lappeenranta, Finland | [2]Department of CSE, United International University, Dhaka, Bangladesh

Correspondence: Md. Saddam Hossain Mukta (saddam.mukta@lut.fi)

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ABSTRACT

Many programming languages and technologies have appeared for the purpose of software development. When choosing a programming language, the developers' cognitive attributes, such as the Big5 personality traits (BPT), may play a role. The developers' personality traits can be reflected in their social media content (e.g., tweets, statuses, Q&A, reputation). In this article, we predict the developers' programming language preferences (i.e., the pattern of picking up a language) from their BPT derived from their content produced on social media. We randomly collected data from a total of 820 Twitter (currently X) and Stack Overflow (SO) users. Then, we collected user features (i.e., BPT, word embedding of tweets) from Twitter and programming preferences (i.e., programming tags, reputation, question, answer) from SO. We applied various machine learning (ML) and deep learning (DL) techniques to predict their programming language preferences from their BPT. We also investigated other interesting insights, such as how reputation and question-asking/replying are associated with the users' BPT. The findings suggest that developers with high *openness, conscientiousness,* and *extraversion* are inclined to mobile applications, object-oriented programming, and web programming, respectively. Furthermore, developers with high *openness* and *conscientiousness* traits have a high reputation in the SO community. Our ML and DL techniques classify the developers' programming language preferences using their BPT with an average accuracy of 78%.

1 | Introduction

Over the years, many programming languages have appeared in different programming paradigms, each serving a niche scope within said paradigm [1]. Therefore, software developers have many choices when picking a suitable language for their project [2]. Developers may write codes in different programming languages. However, the performance of these languages may vary in terms of safety, convenience, simplicity, and response [3]. To optimize software development workflows, improve team cooperation, and deliver high-standard products, it is important to comprehend the psychological traits that might drive a developer to pick up a particular programming language [4].

When facing problems, software developers often rely (i.e., asking a question) on Stack Overflow (SO) feedback and activities [5]. Many expert developers contribute (i.e., answering, voting) to help other developers in the programming community [6]. On the other hand, other social media sites (i.e., Facebook, Twitter, and Reddit) have become an integral part of our daily lives. Users' tweets and posts have become a great source of information for researchers when it comes to predicting their behavior [7]. Several studies [8, 9] have also been conducted to identify users' personality traits from their social media content. Therefore, it might be possible that programming language choices or patterns of community activities are associated with the developers' personality traits and can be derived from their social media usage.

Personality traits [10] describe the pattern of feelings, thoughts, and behavior of an individual. Several studies [11–13] have appeared that predict a user's behavior or actions based on her personality traits. O'Tuathaigh et al. [11] showed that personality factors are strong determinants of a student's behavior as a medical professional. Similarly, personality traits have a strong correlation with individual movie preference [14], mate selection [15], career decision making [16], and many more [17, 18]. In light of the above discussion, we conclude that personality traits might influence user behavior and actions to a significant degree. To the best of our knowledge, this paper is the first to investigate how psychological attributes (Big5 personality traits (BPT) [19, 20] which is predominant among all personality models) influence the user's programming language preferences in real life. To be more specific, no study has investigated programming language preference prediction from their BPT derived from social media content. In addition, it is also important to investigate how the preference prediction is trustworthy and explainable. Furthermore, while question-answering systems such as SO have been extensively practiced among developers, the connection between the users' pattern of question/answering and their psychological attributes remains unexplored.

In this study, we employ a novel strategy by computing the BPT of a total of 820 Twitter (aka X) users from their tweets and extracting their programming preferences from their SO. Then, we built a novel classification model to predict the users' programming language preferences from their BPT derived from tweets. Although we can predict the users' personalities from SO itself, the questions and answers largely reflect the programming context of SO. In contrast, users frequently express their thoughts, ideas, and intimate information on social media [21, 22]. Thus, identifying BPT from social media is likely to produce accurate results. We computed the users' BPT scores by using IBM Watson personality insights API [23]. We also reasonably narrowed down the programming tags into four class labels [24] (oop, web-pro, database, and mobile) that are popular according to SO statistics [25]. We use our ML/DL-based approach with explainability of the models to predict the users' programming language preferences from their tweets. Furthermore, we found that the users' different online community activities (question asking, answering, etc. [26]) in SO are strongly linked to their BPT.

In summary, we investigate and answer the following research questions (RQs):

- RQ1: How is BPT associated with the developers' programming language preference?
- RQ2: To what extent can the developers' programming language preferences be predictable and explainable from their BPT?
- RQ3: How are BPT and developers' community behavior (i.e., questioning, answering, and reputation) linked with each other?

Through a combination of predictive modeling and empirical analysis, our effort aims to develop a framework for predicting programming language preferences based on BPT. To sum up, we have made the following contributions. First, we cross-link Twitter and SO social networking sites to predict the users' programming language preferences from tweets. Instead of manually collecting questionnaire-based programmers' preferences, our study collects the developers' recent choices from their digital footprints by cross-linking these two sites. Second, we identify an interesting association between the users' BPT and their preferences in terms of programming languages. We further explore the association between the users' questioning, answering, and reputation scores. Moreover, we analyze whether the Twitter and SO content reveal similar BPT. Finally, we built machine learning and deep learning-based prediction models to determine the developers' programming language preferences from their BPT with the explainability of the feature selection. Developing a prediction model for programming language preferences and the analysis of community activities based on the developers' BPT may have many real-life applications. For example, recruiters, job agencies, and management may develop a tool that facilitates employing project-specific developers by observing their tweets. Managers may also anticipate the developers' ability to perform group activities (i.e., problem-solving skills by observing their answering pattern in SO and reliability in the team by investigating high reputation in the context of SO).

An overview of the section organization is as follows. Section 2 describes the background and related work. Sections 3 and 4 are all about the dataset collection and balancing of the dataset. Section 5 conducts the analyses of the research questions, and Section 6 contains the implications and discussion. Section 7 presents the concluding remarks.

2 | Background and Related Work

In recent years, question and answer (Q&A) websites have gained popularity for their accessibility, diversity, and high-quality content. They hold the attention of a wide range of audiences, from researchers to general users. Quora which is one of the biggest Q&A websites, has 300 million unique users monthly. Stack Overflow (SO) is another acclaimed Q&A website in the developer community, where almost 6300 questions are being asked per day. Since SO posts are written in natural language, comprehensibility metrics and textual context analysis tools can potentially provide us with valuable information on what makes a posting go on to be perceived as trustworthy by the community.

Big5 Personality Traits (BPT): Goldberg et al. [27] described how BPT has five different traits: *openness to experience, conscientiousness, extraversion, agreeableness*, and *neuroticism* as illustrated in Figure 1. Openness encompasses a variety of interest and information-seeking behaviors, as well as intellectual curiosity, inventiveness, and originality [28]. Park and Kim [29] discussed that individuals with high conscientiousness are typically

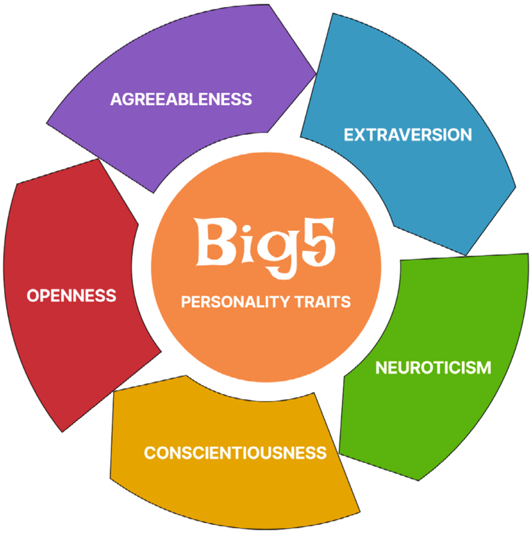


FIGURE 1 | Big5 personality traits (BPT) proposed by Goldberg [27].

responsible, cautious, patient, well-organized, focused, meticulous, and achievement-oriented. Learning practices with goals in mind are referred to as conscientiousness as well. Positive feelings and sociability are traits of extraversion.

Extraversion is the personality attribute that has been linked to a variety of leadership qualities [30], like being polite, adaptable, trustworthy, amiable, cooperative, forgiving, soft-spoken, and tolerant are examples of agreeableness. The tendency to be altruistic—sympathetic, kind, humble, value-aligned, and conflict-averse—is represented by agreeableness [31]. Individual traits associated with neuroticism include emotional instability, impulsive control, and a disposition toward unpleasant emotions.

Automatic Personality Recognition (APR): Two major drawbacks of self-reported BPT are as follows: (i) subject tends to idealize (i.e., better than average) himself by responding to a biased answer [32] and (ii) subject likely responds to the reference-group [33] impression of those who remain in his surrounding environment. For example, although an individual is introverted in reality [34], he considers himself an extrovert since his companions from the surroundings likely represent extrovert behavior. Several studies [35–37] have been conducted on APR from social media content. For example, Golbeck et al. [38] determined users' personalities from their tweets using BPT. The authors developed a Twitter app with 45 questions from BPT to observe the users' initial personality traits and later recruited 50 users for the final test. Afterwards, the tweets were analyzed using a psycholinguistic text analyzer called Linguistic Inquiry and Word Count (LIWC) [39]. Gaussian Process and ZoreR regression analysis were performed to predict the users' personality scores. By analyzing their social media data, Quercia et al. [40] categorized the Twitter users based on their BPT scores. The users were divided into listeners (follow many users), popular (followed by many users), and highly-read (often listed in other reading lists) categories based on their number of followers and following listed counts. Their study shows that listeners and popular users generally have high extraversion and low neuroticism scores. Moreover, influential users are more organized about their lives, thus, they have a high value of conscientiousness.

In a recent study, Dhelim et al. [23] presented that IBM Watson Personality Insights can develop an APR using the users' digital footprints such as email, text messages, tweets, and forum posts. After considering the downside of self-reported personality tests and the difficulty of conducting a personality test among anonymous users, we were motivated to use IBM Watson Personality Insights APR to compute the individual BPT scores.

Impact of BPT on Users' Behavior: Gnambs [41] discusses about different personality traits and their correlation with programming aptitude. Among the personality traits, openness is the most important for writing successful programs. Programmers who are conscientious are less prone to making errors while programming. Moreover, extraversion also acts as another contributor to successful programming execution. However, the authors did not find any association between agreeableness and neuroticism traits with programming aptitude. Salleh, Mendes, and Grundy [42] found there to be an impact due to BPT, that is, openness, conscientiousness, and neuroticism, on the effectiveness of pair programming (PP). Similarly, Amin et al. [43] investigated the BPT and knowledge-gathering behavior, which have an effect on a programmer's creativity. The study also shows that neuroticism has a negative effect on creativity, while openness, extraversion, and conscientiousness traits have a positive association with it. Hannay et al. [44] explored the relationship between task complexity, skill, and the BPT on pair programming performance. Their findings highlight the possibility of understanding programming language preferences using multiple factors such as programming expertise, collaborative dynamics, and so forth Yilmaz et al. [45] found that using an interactive personality profiling approach may facilitate efficient software team structuring. Zhou et al. [18] discovered that personality traits can predict everyday behavior, such as an individual's reading preferences and purchasing tendencies. Wuertz [17] found an association between the users' BPT and their scale of concern regarding environmental awareness.

Behavior Analysis in SO: Several studies [46–48] have also appeared in the literature seeking to understand user behavior in SO. Mustafa, Zhang, and Naveed [46] explored what motivates users to contribute to online communities like SO. Based on cognitive and social exchange theory, they found that the reciprocation of knowledge positively influences contribution, while knowledge seeking influences knowledge contribution negatively. Rubaye and Sukthankar [47] examine how reputation influences user interactions in tech-related online Q&A communities. The study explores the correlations between the author's reputation and the user's commenting behaviors, highlighting how comment timing is affected by the commenters. Zolduoarrati, Licorish, and Stanger [48] showed that the users' contributions on online Q&A sites like Reddit and SO for the purpose of global knowledge exchange is associated with their gross domestic product (GDP), location, and urban infrastructure.

AI Research in SO: We also found several studies [49–52] that have applied AI techniques over the SO dataset to predict the users' activities. Jiang et al. [49] presents a method with fully convolutional neural networks for predicting questions on SO, showing the dynamics of user engagement and predicting content popularity. Roy et al. [50] explored how both deep learning (DL) and machine learning (ML) research contribute to

TABLE 1 | Comparative analysis between our proposed framework and related studies.

Authors	BPT vs. programming pref.	Twitter & SO fusion	BPT vs. Repu.	BPT vs. Q/A
Hannay et al. [44]	×	×	×	×
Salleh, Mendes, and Grundy [42]	×	×	×	×
Rahman et al.	×	×	×	×
Amin et al. [43]	×	×	×	×
Bazelli, Hindle, and Stroulia [53]	×	×	×	✓
Papoutsoglou et al.	×	×	×	✓
Our proposed	✓	✓	✓	✓

community question-answering (CQA) websites such as Yahoo! Answers, Stack Exchange, and SO. He et al. [51] explored the necessity of robust SO post representation models, where BERTOverflow has a weaker performance than Post2Vec in different downstream tasks. The authors later introduce SOBERT, a pre-trained representation model with SO text that outperforms the other models in various downstream tasks. Raida et al. [52] examined the difficulties faced by SO users in raising unclear and complex questions, which frequently leave questions unanswered. They evaluated the complexity of the question using TF-IDF, LDA, and Doc2Vec models, where the Doc2Vec model performs better.

However, in the light of the above discussion, we found that our study first investigated the following aspects: (i) finding a novel fusion between two social media sites (i.e., Twitter and SO) to retrieve interesting information, (ii) computing the association between BPT versus the developers' programming preference and building a machine learning model with explainability, and iii) exploring the relationship of the developers' BPT with their community behavior such as questioning, answering and reputation in SO. Table 1 presents the comparison between our study and prior studies in terms of the different dimensions.

3 | Dataset Collection

We collected the users' information randomly from SO by searching for questions with the most popular tags. We also searched for these users on Twitter using the same *username* as on SO. We found 962 users with the same username on both Twitter and SO. We confirmed it by observing the users' profile photos, common shared websites, professional affiliations, and locations. We also noted that 55 users were inactive on Twitter, and the tags of 87 users did not match our class labels. Thus, we discarded the data of these users due to the lack of adequate information for further analysis. Our final sample size reached a total of 820. Then, we extracted their programming preferences from SO using the Python *Stack Exchange API* implementation package.[1]

We collected data on a total of 1,494,072 users' tweets using the Python *tweepy* implementation package. Table 2 presents the statistics of our Twitter dataset. Then, we computed the users' BPT scores using *IBM Personality Insight API*. Arnoux et al. [54] showed that *IBM Personality Insight API* outperforms state-of-the-art techniques for personality computation. After computing the users' BPT scores, we kept this data as the independent variables and the users' 3 best SO tags as the dependent variables [24].

TABLE 2 | Statistic of our Twitter dataset.

No. of users	820
Total number of tweets	1,494,072
Average number of tweets	1,835.469
Maximum number of tweets of a user	3,247
Minimum number of tweets of a user	9
Total number of words in tweets	21,962,414
The average number of words in tweets	26,980.853
Maximum number of words in a tweet of a user	42,123
Minimum number of words in a tweet of a user	68

We observed that different users apply different tags for the same language. For example, c++11, c++14, c++17 tags are used for c++. We removed these tags and replaced them with the main programming language, i.e., c++. When we merged the related programming languages into a single class, the dataset showed interesting patterns that gave a thorough picture of developer preferences in the software development environment.

We observed that javascript and C# are quite prevalent, which are positioned as the first and second most popular languages, respectively [55]. On the other hand, Python has obtained huge popularity and extended its application across multiple domains [56]. Figure 2 presents a timely representation of programming language preferences, which is developed from our dataset and also indicates the recent trends in the software industry [57].

Table 3 shows that we have a total of 1,908 tags (138 distinct tags) for a total of 820 users. We found that oop has the highest number of tags (i.e., 900), while the database has the lowest number of tags (i.e., 116). We categorized the languages with similar structures, syntax, and capabilities using broader class labels. For example, *.net* has relevance aligned with *C#, f#, asp.net, vb.net, and xml.* These languages use a *Microsoft* backed object-oriented platform; we also labelled these languages as object-oriented programming (oop) along with mainstream oop languages. Later, we categorized several tags such as *php, ruby, Perl, jquery, javascript, node.js, angular js, react js and typescript, html, CSS, JSON, django* as the web-pro (web programming) class. We reasonably made broader categories of four different class labels: *oop, web-pro, database,*

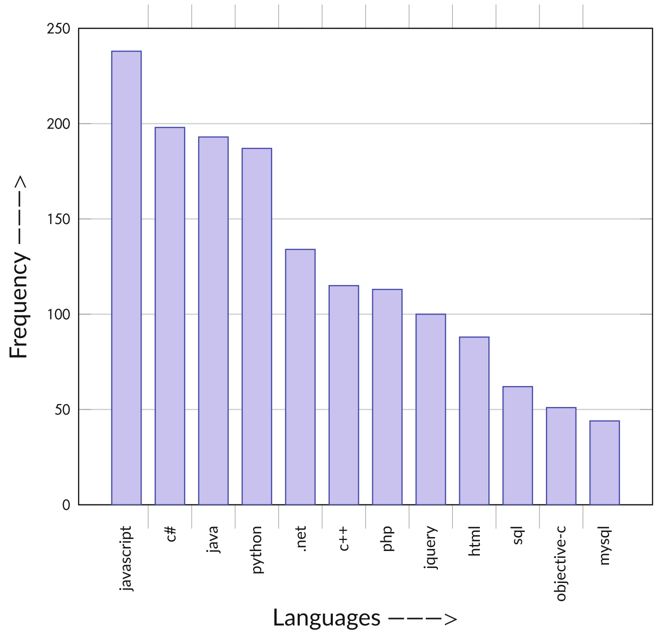


FIGURE 2 | Distribution of the programming language tags used by the developers in our dataset.

TABLE 3 | Different class labels, tags, and number of instances in the SO dataset.

Class label	Related tags	Instances
Oop	.net, c#, f#, asp.net, vb.net, java, python, swing, c++	900
Web-pro	php, ruby, perl, jquery, javascrpt, node.js,angularjs, reactjs, typescript, json, django	670
Mobile	kotlin, objective-c, swift, android, ios	222
Database	sql, sql-server, mysql, oralce, postgreSQL	116

and mobile from the most popular tags. Model predictions are easier to understand and comprehend when the classification challenge is simplified [58]. Rudin [59] also presented that machine learning models become more interpretable, efficient, and effective when the class labels are minimized. We also extracted the users' SO comments and answers for further textual analysis. We collected a total of 24,540 and 16,510 answers and comments, respectively. We found an average of 30 and 20 answers and comments, respectively, for each user.

4 | Balancing Dataset

Table 3 shows the data distribution of the different classes in our dataset. However, the table shows that our dataset has a class imbalance problem [60]. The class distribution for oop, web-pro, mobile, and database has 900, 670, 222, and 116 instances, respectively. We addressed the class imbalance problem using the Synthetic Minority Oversampling Technique (SMOTE) [61]. By using SMOTE, we created artificial instances for minority classes in order to balance the distribution of classes in the dataset. SMOTE is an oversampling approach [62] that compensates for the imbalance problem without bias in favor of the dominant class by producing synthetic samples. We use the Python imbalance-learn implementation package to balance our dataset.

Figures 3 and 4 are two bar charts showing the visual representation of the class labels before and after applying the SMOTE technique [24]. Table 4 gives a summary of the distribution of the dataset both before and after SMOTE. We obtained a balanced distribution of instances among the various classes, which can enhance the performance of machine learning models.

5 | Analysis of the RQs

5.1 | RQ1: How is BPT Associated With the Developers' Programming Language Preference?

- **Motivation:** The developers' BPT likely influences their problem-solving skills, work habits, and overall preferences in their working environment. For example, the openness trait may have a correlation with mobile platform-based programming languages. Trabucchi and Buganza [63] discussed that mobile app technology is producing a growing number of significant inventions based on digital technologies (i.e., cryptocurrency wallet, unity-game-engine, virtual-reality, augmented-reality, and kotlin-coroutines) due to the widespread usage of smartphones. Similarly, developers who are responsible, patient, and goal-oriented are likely to be interested in OOP-based programming languages. The maintainability of programs can be enhanced when a language fully adopts OOP notions (i.e., abstraction, encapsulation, and polymorphism) [64]. Conscientiousness might encourage developers to maintain and organize the code for reuse and to be easy to understand further. On the other hand, developers who have high extraversion scores may have a correlation with web programming (web-pro). Studies [65, 66] discuss that an e-commerce developer might need to be aware of the latest technologies (motion UI, serverless framework, WebSockets, static site generators, jam stack, etc.) and understand well how different organizations collaborate to create and promote their own e-commerce profiles. Extraverts tend to be outgoing and perform well in situations requiring a lot of conversation and engagement. Developers in web development projects are likely to share content, outcomes, tips, and tricks because of their tendency to collaborate with others. However, neuroticism traits may not have a significant correlation with any platform-based programming language since negative traits may reduce working motivation [67, 68].
  In light of the above discussion, we also found relationships between the developers' pattern of tweets and their SO tags, which can be relatable to their BPT. We observed that *User_X*'s tweet below exhibits a high degree of conscientiousness since it shows concern about issues in society, like the climate crisis, and demonstrates a sense of responsibility. The user likely assigns primary tags in SO that fall under the label of oop.

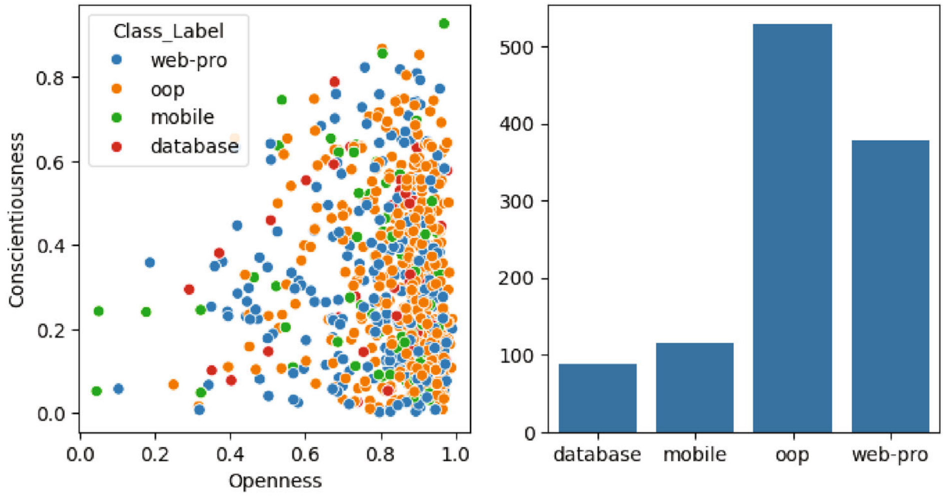


FIGURE 3 | Class label distribution before using SMOTE.

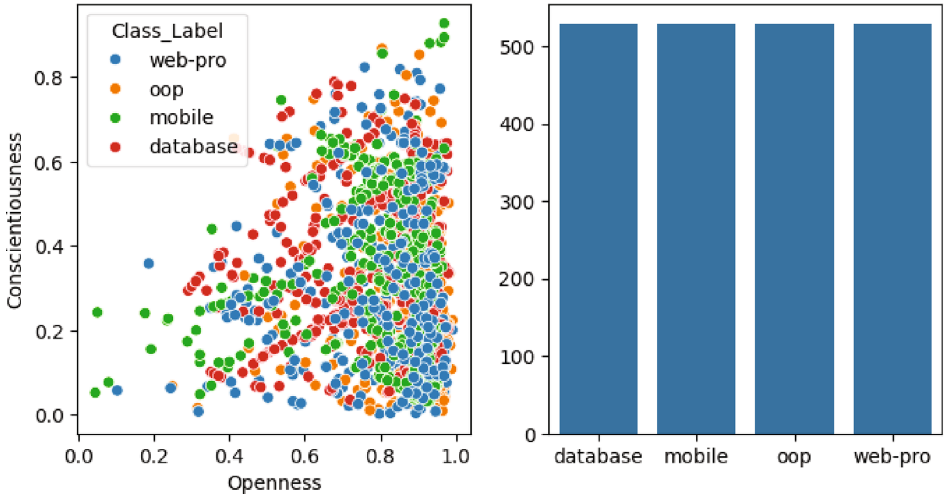


FIGURE 4 | Class label distribution after using SMOTE.

TABLE 4 | Data distribution before and after oversampling.

Class label	Before SMOTE	After SMOTE (% of oversampling)
Oop	900	0%
Web-pro	670	134.33%
Mobile	222	405.40%
Database	116	775.87%

Conscientiousness vs. oop
User_X
Stack Overflow Tags: c#, .net, entity-framework, asp.net, etc.
Tweets: This is what keeps me awake at night. Coding and IT security are interesting problems. But what really scares me is the climate crisis. It's here. Now.

The tweet of *User_Y* reveals a great degree of excitement to learn new experiences because the tweet expresses interest in taking part in an intellectually challenging task. Similarly, the user also shows interest in SO with the tags of mobile application development.

Opennes vs. Mobile
User_Y
Stack Overflow Tags: objective-c, ios, iPhone, Xcode, etc.
Tweets: #adventofcode starts tomorrow! I hope you've all got your projects set up and ready to go. Good luck! May you get all 50 stars!

Similarly, *User_Z* praises his/her friend's creation by tweet. By sharing his/her excitement, the user hopes to establish a connection with their social circle and recognize a friend's accomplishment. This post demonstrates extraversion traits by expressing happiness and engaging in social interactions. The user's SO tags represent their preferences for programming tags that are associated with their BPT.

Extraversion vs. Web-pro
User_Z
Stack Overflow Tags: angularjs, javascript, html, jquery, etc.
Tweets: Really like the finished product. My friend Bruce made these cool bellows for his outdoor fireplace.

- **Approach:**
  Feature Extraction and Selection: In the light of the above discussion, first, we found important features that are correlated between the users' BPT and their programming language preferences. Later, we applied conventional machine learning (ML) and deep learning (DL) techniques to understand to what extent the users' programming language preferences can be predicted from their BPT.
  Feature Selection From BPT: We conducted a multi-level discriminant analysis [69] using *SPSS* where BPTs were the predictors and the users' programming language preferences were the predicted (i.e., dependent) variables. The BPTs computed from *IBM Personality Insight API* are *openness, conscientiousness, extraversion, agreeableness, and neuroticism*. For selecting important personality traits, we used *Fisher's linear discriminant analysis (LDA)* [70], which found a correlation between independent variables and dependent variables having more than two class labels. Fisher's LDA function scores were proportional to the coefficients of multiple linear regression with the dependent variables where larger (>1.0) score predictors produced more accurate outcomes [71].
  Feature extraction from BERT: We also extracted features using bidirectional encoder representations from transformers (BERT) over our tweets [72]. BERT (768 dimensions) captures word meanings in context and understands word semantics based on the surrounding words. BERT also handles polysemy (multiple meanings of words) and captures nuances in language better.

- **Results:**
  Table 5 shows the Fisher's coefficient between the users' BPT and their programming language preferences. We found that our best-selected features were *openness, conscientiousness* and *extraversion* traits. Openness has a stronger correlation (61.319) with mobile platform-based programming languages than OOP, web-pro, and the databases. Conversely, the conscientiousness trait has a higher correlation value (42.149) with OOP than mobile platform-based programming languages. Conscientious developers are likely to work with OOP (python, java, c#, and C++, etc.). However, the extraversion trait has a higher correlated score (18.370) with web-pro than other platforms. However, we found no correlation with any BPTs and database tags. Since the majority of developers (i.e., oop, web-pro, mobile) need assistance with the database platform, they also joined the conversations with other threads. Fan et al. [73] stated that when learning in open-ended environments, everyone should have knowledge about databases. Therefore, we perhaps did not find any specific correlation with any one BPT. However, developers with neuroticism traits have no correlation. In [41], the authors found a similar association to ours (i.e., openness and conscientiousness) when finding a correlation with programming aptitudes.

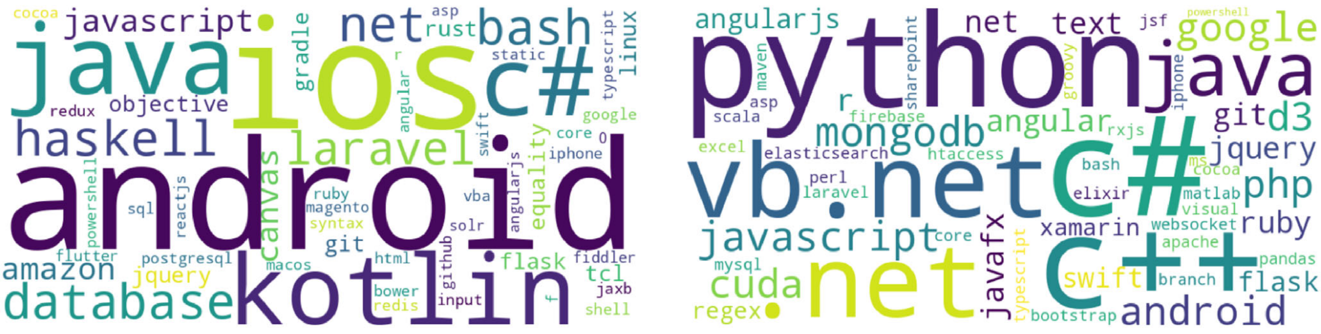
Developers' BPT and Their Topical Interests: We also found insights among the participants' involvement patterns in SO based on their BPT. By combining user-generated tags based on personality traits (especially openness and conscientiousness), we found a pattern by utilizing data visualization techniques such as word clouds [74]. We separated the users who had scores for their openness and conscientiousness traits over the 3rd quartile and applied the word cloud visualization for a deeper understanding of the underlying reasons and preferences. Figure 5 presents the relationships among the user-generated content, personality traits, and SO tags. For example, we found that users with high openness scores have the most popular tags, such as android, python, java, and so forth, in SO, which are also endorsed by our discovered association. Figure 5a,b show that the users have higher openness and conscientiousness scores, respectively.

5.2 | RQ2: To What Extent Can a Users' Programming Preferences Can Be Predictable and Explainable From Their BPT?

- **Motivation:**
  Jiang, Gradus, and Rosellini [75] showed that ML can be applied for complex multi-modal assessments for psychological traits and disorders. However, traditional ML techniques can only build a prediction model. In recent times, researchers have shown less interest in accepting the outcome of these traditional models due to a lack of explanation of the nuances and causality [76, 77]. To address this issue, it

TABLE 5 | Fisher's LDA function coefficient between the users' Big5 traits and their programming preferences.

	Oop	Web-pro	Database	Mobile
Openness	53.673	33.982	30.890	61.319
Conscien.	42.149	21.799	21.272	28.255
Extra.	−0.037	18.370	6.737	−1.907
Agree.	0.674	0.784	0.329	0.410
Neuro.	0.069	0.292	0.591	0.521



(a) (b)

FIGURE 5 | Word cloud showing the topic of interest by openness and conscientiousness personality traits. (a) Word cloud for Openness (scores above 3rd quartile). (b) Word cloud for Conscien (scores above 3rd quartile).

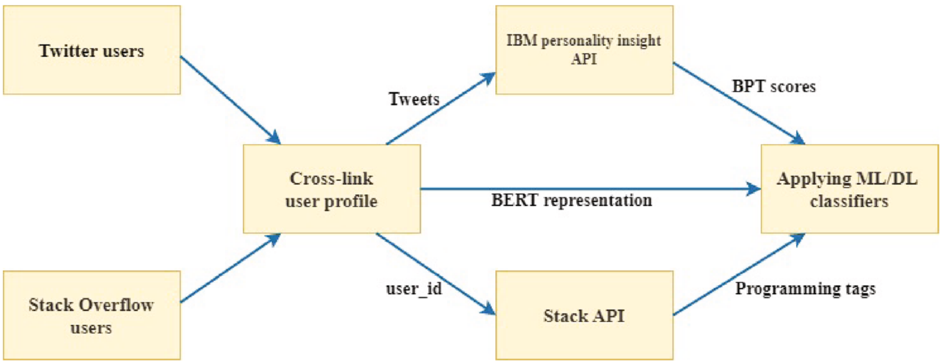


FIGURE 6 | Pipeline of our programming preference prediction framework from BPT.

is also crucial to incorporate the explainability of our models along with the model's performance based on the importance of different features [78, 79].

- **Approach:**
  Building machine learning models: In this section, we built a classification model to predict programming language preference. We randomly split the dataset by 80% and 20% for training and testing, respectively. We outlined the methodology and steps undertaken to construct the predictive model as shown in Figure 6. The tags have been categorized into four different classes (see Table 3). We consider the following classification algorithms in our modeling approach:
  - *Support Vector Machines (SVMs)*: SVMs are versatile and effective for binary and multi-class classification tasks [80]. In our case, we leveraged SVM to address the task of multi-class classification, where each observation can belong to multiple classes simultaneously. The SVM model was trained on our dataset, which included feature vectors derived from BPT scores [81], with each observation associated with one or more programming tags as labels.
  - *Random Forest (RF)*: RF is a powerful ensemble learning technique that has gained widespread popularity in the field of ML due to its ability to deliver robust, accurate, and versatile predictions [82]. Its versatility and outstanding performance make it a formidable choice for a broad spectrum of real-world applications.
  - *K-Nearest Neighbors (KNN)*: KNN classifier is built with five neighbors. The model is trained and then applied to predict outcomes based on unobserved data. This model was implemented using the scikit-learn implementation package in Python. The model makes use of the KNN algorithm's capacity to classify the data points according to how close they are to other points in feature space in order to precisely assign class labels to input instances [83].

  Building Deep Learning Models: We also applied deep learning methods by using features extracted from BERT embeddings and BPT with convolutional neural networks (CNN) and Feedforward Neural Networks (FFN) [84]. These advanced frameworks offer improved performance for classification tasks.
  - *CNN Model Architecture*: In our model, we employed 1D CNN models for text classification tasks and they showed promising results for identifying complex patterns within text sequences [85]. Because pooling layers offer translation invariance, the model can concentrate on the existence of particular characteristics rather than their exact placement [86, 87]. We present our CNN model architecture in Figure 7. The details of the deep learning architecture are as follows: two one-dimensional convolutional layers with a filter size of 64 and a kernel with a size of 3 rows and columns, which make up the CNN model. A one-dimensional max-pooling layer follows each of these layers. These layers extract features from the input vectors. We used a fully connected neural network layer with 128

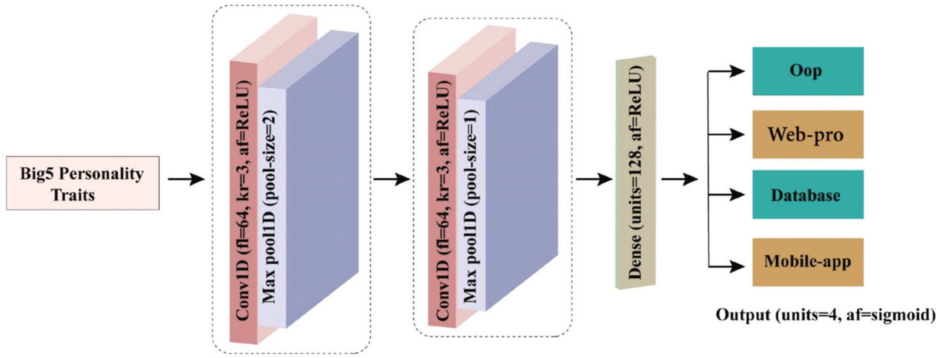


FIGURE 7 | 1D CNN architecture of our framework.

TABLE 6 | 1D CNN architecture for our experiment.

Layer	Hyperparameters
1D convolution	fl = 64, kr = 3, af = ReLU
1D Max-pooling	Pool-size = 2
1D convolution	fl = 64, kr = 3, af = ReLU
1D Max-pooling	Pool-size = 1
Dense	Units = 128, af = ReLU
Output	Unit = 4, af = sigmoid

Abbreviations: af = activation function, fl = filter size, kr = kernel size.

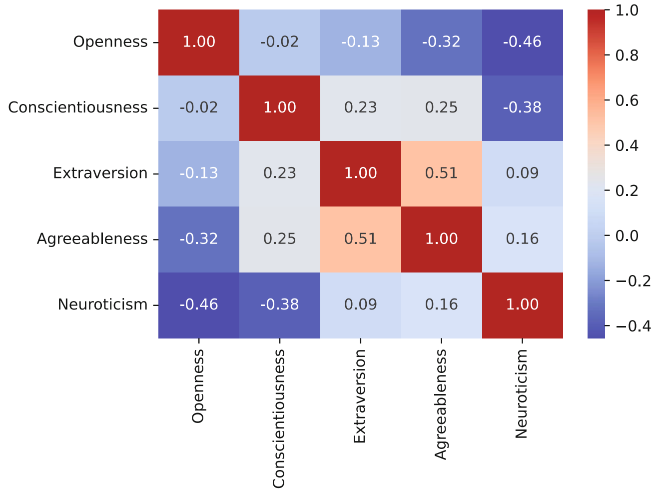


FIGURE 8 | Correlation among the personality traits.

neurons to convert the retrieved features. After that, we predicted the input sequence labels using a sigmoid [88] output layer. We used the activation function ReLU [89] for all other layers. Table 6 summarizes our model parameters and configurations.

- *FFN Model Architecture*: We also applied the FFN model, which consists of four layers:
  - Input Layer (Dense Layer): This layer took BERT embedding as the input data and adopted the ReLU activation function [90], which introduces non-linearity, enabling the model to learn complex patterns in the data.
  - Output Layer (Dense Layer): This layer has a softmax activation function [91] in the output layer with four neurons, producing the multiclass classification prediction. There were two hidden layers except for the input and output layers. The model was compiled using an *adam* optimizer. The training process was monitored using validation data to assess the model's performance over 100 training epochs.

- **Explainability of the BPT-Based Model:** To better understand how our model predicted the programming preference based on the BPT, we applied the SHAP Python implementation package,[2] which enabled us to look into the contribution of the feature sets. Instead of dealing with the feature sets as black boxes, it was important to understand the reasoning behind why the models performed as they do. Explainability is crucial to ensure that the model is not biased and behaves rationally [92].
  First, we present the correlation among the personality traits themselves in Figure 8. We found that openness and neuroticism traits are negatively correlated; individuals with high scores in openness embrace change and challenges, whereas those exhibiting high scores in neuroticism tend to display excessive fear and hesitation in their decision-making processes. Conversely, we found that extraversion and agreeableness traits show a positive correlation with each other. Extrovert individuals are likely to be skilled at performing social interactions, and these qualities may have overlapping patterns with the traits of agreeableness. These findings are also compatible with the previous literature [93], which might be relatable to the context of programming preference prediction. As our model is a multiclass classifier, we further investigated how the model makes decisions for each of the individual classes. Utilizing the Shap Library (SL), graphical representations have been rendered in logit space to explain these processes. Figure 9 shows Bee Swarm charts to present our model explainability. The distance from the normal line in the middle shows how strongly the trait influences the prediction. The red points on the right and the blue points on the left indicate positive correlations, whereas the opposite suggests a negative correlation. We can observe from the figure that developers with the openness (strong) and conscientiousness traits have a positive correlations with oop. On the other hand, web-pro developers tend to show low consciousness and higher extraversion traits. For mobile, the developers

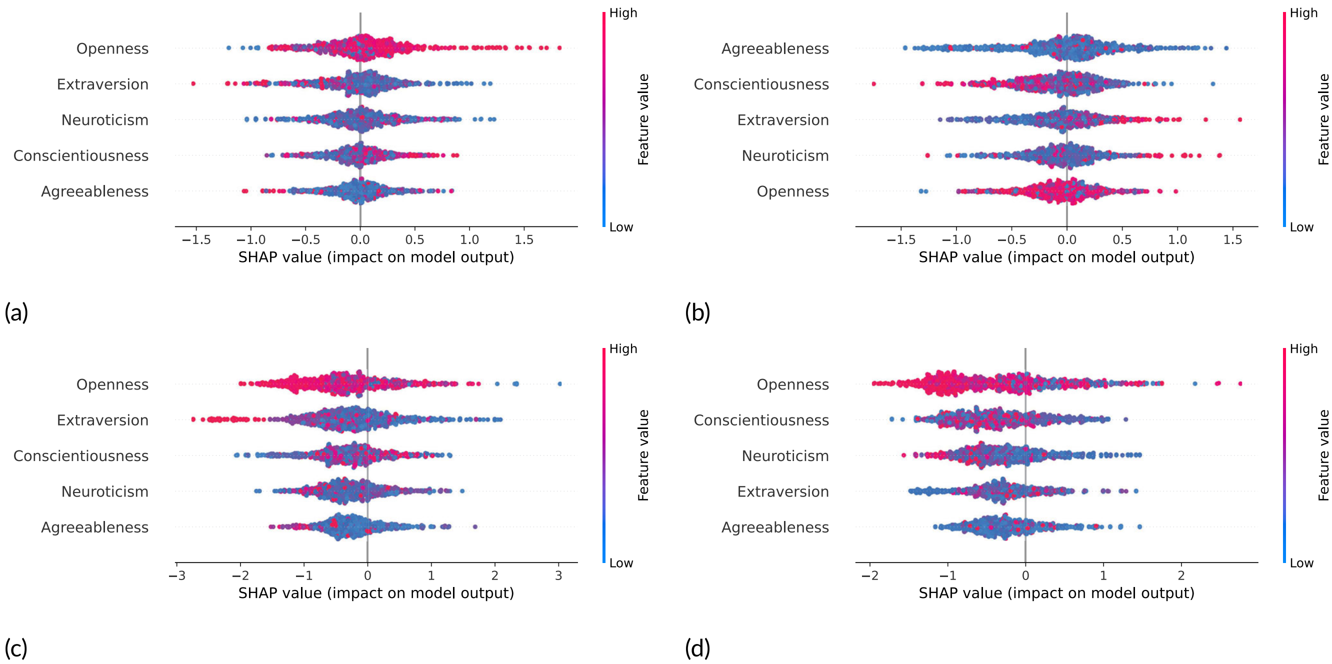


FIGURE 9 | Contribution of BPT in the prediction of programming preferences. (a) oop, (b) Web-pro, (c) Mobile, (d) Database.

TABLE 7 | Performance of the BPT-based prediction of programming preference.

Classifier	Precision	Recall	F1-Score	Accuracy
SVM	0.78	0.73	0.78	0.78
RF	0.65	0.63	0.65	0.66
KNN	0.62	0.60	0.61	0.62
1D CNN	0.76	0.73	0.73	0.76
FNN	0.75	0.73	0.75	0.75

show a higher correlation with the extraversion and openness traits. The developers dealing with databases have a negative correlation to openness; they perhaps tend to perform routine work with less variation in their tasks. The database system may be used in the majority of the programming platforms, therefore no specific correlations are observed.

- **Results:** Table 7 presents the performance of our prediction model using traditional ML models, that is, SVM and RF, using our BPT features. Then, we determined the accuracy of 78% using the SVM model. On the other hand, we found the word embedding-based 1D CNN and FNN models have an accuracy of 76% and 75%, respectively.

5.3 | RQ3: How Are BPT and the Developers' Community Behavior Are Linked With Each Other?

As developers navigate platforms like SO and Twitter and interact with the community, their posts and responses are likely to reflect their behavioral traits [94, 95]. In SO, most of the interactions are through questions and answers that represent the users' technical interests. Twitter is more of a social platform where people post their personal experiences and thoughts, intimate personal information, daily activities, and their observations on any issue. Studies [38, 96] show that users are likely reveal their actual personality traits on social media. Due to the nature of these two platforms, the method of interaction with the respective communities is expected to be different. In this section, we have explored several objectives to address the RQ3.

- **Objective 1 (O1):** Does BPT affect a user's propensity to engage in question-and-answer interactions on SO?
- **Objective 2 (O2):** Does a user exhibit the same BPT across Twitter and SO?
- **Objective 3 (O3):** Do the BPTs have any effect on the community behavior of a user in SO?

Observations:

O1: We explored the association between BPT and the question-and-answer pairs. Table 8 shows the association between questions and answers for different personality traits. Openness and conscientiousness are two notable personality traits that show significant associations with the answer that a user provides.

On the other hand, the connections between the answers and BPT, such as extraversion, agreeableness, and neuroticism, did not achieve statistical significance. However, we did not find any correlations between BPT and asking questions in SO. These findings raise questions about the complex and multidimensional characteristics of personality evaluation in the context of community interactions.

We further explored the association between the developers' BPT and their SO answers/comments and conducted an exploratory

TABLE 8 | Association with BPT and question asking/answering.

Big5 personality traits	*p* (question asking)	*p* (answering)
Openness	0.4396	**0.0046****
Conscient	0.5449	**0.0217***
Extravert	0.46598	0.111
Agreeable	0.1623	0.355968
Neuroticism	0.373679	0.829

Notes: Statistically are correlated (in bold). Significant correlations are shown in * where $p < 0.05$.

analysis of the textual data. The analysis showed there to be a few intuitive topics from the users' comments and answers, which are relatable as to why developers show a propensity to interact in SO. To perform the analysis, we applied different topic modeling techniques [97]. In this regard, we performed the following steps. In preprocessing, we tokenized the input text into sentences, expanded any contractions, removed stop words and unwanted characters, tokenized words, and performed lemmatization. Then, we selected users with high BPT scores for different traits (openness, conscientiousness, and extraversion). Since these traits correlate with the users' programming language preferences, we performed a textual analysis of their SO comments and answers. We selected the users with each trait with a high BPT score above the third quartile range. Later, we selected the Latent Semantic Analysis (LSA) and Latent Dirichlet Analysis (LDA) topic modeling techniques to apply to the dataset. LDA is a generative model that explicitly identifies topics and their distribution over the users' comments. In contrast, LSA finds hidden patterns and captures semantic relationships among the topics [98]. Since we were looking for semantic relationships among the topics, we applied LSA using the Gensim[3] Python implementation package. We found a correlation between the pattern of comments and answers of SO users and their high openness traits. These users generally remain active and perform challenging tasks. For example, users with high openness scores are likely to solve problems and find alternative solutions to existing issues. They are also prone to applying different techniques and replying to users with multiple solutions. We also extracted topics from the SO comments and answers of the users with high conscientiousness scores. We found some topics that were relatable to the users' meticulous characteristics in SO. For example, these users are careful, effective, and well-organized in their coding practices and standards during interactions in their software development ecosystem. Finally, we also observed that extravert users are prone to discussing the topics more to clarify the answer seeker's question. They also shared the ideas they use and plan to work. These findings are intuitive and support previous studies. Lambregts [99] showed that users are likely to be willing to share their knowledge with the community. Duffy and Chartrand [100] showed that extroverted people build rapport with others for the community's well-being.

O2: In this observation, we conducted a *Paired t-test* [101] to check whether the means of the two BPT scores computed from the text of two different platforms (i.e., Twitter and SO) were significantly different using the Python *stat* package.[4] Table 9 shows that all BPTs are significantly different from one platform to another.

O3: In SO, we observed three major community behaviors: question, answer, and reputation. Questions are one of the primary methods by which users seek assistance, and answers are the way of responding to the questions [102]. The users' reputation is a score that is essentially calculated based on the number of votes they earn for participating in any of the three main software engineering (SE) community activities: asking, answering, and modifying questions/answers. In this objective, we explored the association between these three behaviors and with different BPTs.

- **Reputation vs. Question:** The relevancy and quality of an asked question may influence an individual's reputation score in SO. In our study, we computed the *p* value between the users' reputation score and the number of questions, which was *-0.0227** which is significant (<0.05) and negatively correlated. A user with a good reputation is more likely to be asked a question and answer correctly. On the other hand, Cojuharenco and Karelaia [103] showed that the questioners may face competence penalties and high humility because most practitioners do not ask questions at every opportunity, and the questions reveal what the seeker does not know. Calefatoa, Lanubile, and Novielli [104] mentioned how a user's reputation can influence the likelihood that their questions will be answered successfully in SO. They discussed that user-generated questions that are successful should be brief and incorporate code samples into the question body.
- **Question vs. Answer:** We found no direct correlation between the users' number of asked questions and their responses to the answers. However, Zhu et al. [105] found that both askers and answerers actively participate in these discussions, and that the more answers there are, the more likely the participation increases. Our experiment endorses the above findings where the majority of the threads have 15-20 questions and answers the same. Finally, Wang, Chen, and Hassan [106] also discussed that there is a significant relationship between the number of answers asked to a question and the thread getting slower for answering.
- **Reputation vs. Answer:** Highly-reputed users are regarded as more informed and experienced, and their responses are frequently given greater weight and consideration. Tausczik and Pennebaker [107] demonstrated how consistently a user's reputation relates to how well their response is received. In our experiment, we found that the association between the users' answer and their reputation scores was significantly correlated (*p* value is 0.001114*, which is <0.05). Users with high reputation scores are likely to have their responses regarded more seriously by other users in SO. We additionally conducted a textual analysis of the highly reputed users' answers and comments to understand why they tend to attract more user interaction and readership. We applied an unsupervised extractive method with topic modeling using LDA. First, we used reputation as the key component and selected highly reputed users (# of users is 90) with scores above the third quartile of all our dataset's users. Later,

TABLE 9 | Results of the paired t-test on BPT between the Twitter and SO platforms.

Big5 personality traits	Mean stack.	Mean Twit.	t	p
Openness	0.992	0.824	-31.977	3.35E-146*
Conscient.	0.520	0.311	-29.505	7.07E-131*
Extravert.	0.127	0.153	5.503	4.99E-08*
Agreeable	0.003	0.053	16.598	1.52E-53*
Neuroticism	0.114	0.401	37.486	8.75E-180*

Notes: Statistically are correlated (in bold). Significant correlations are shown in * where $p < 0.05$.

we applied comprehensive preprocessing techniques [98] to the comments and answers of highly reputed users. Then, we vectorized the text using the word2vec model, which captures information about the meaning of the surrounding words and feeds the vector into the LDA model to identify topics. We obtained a coherence score of 0.65, which measured our dataset's best topics.

The score indicates that the topics extracted from the data are meaningful and accurate for analysis. Figure 10 presents the best topics extracted from the experiment. According to the topics, users with high reputation scores discuss programming techniques (i.e., code, notebook, git) and assessments (security, commit). Moreover, these topics strongly focus on programming languages (i.e., python), reflecting the questions, queries, and issues (i.e., answers) that users generally encounter. Their skills help others (i.e., share) understand various programming environments and address challenging issues.

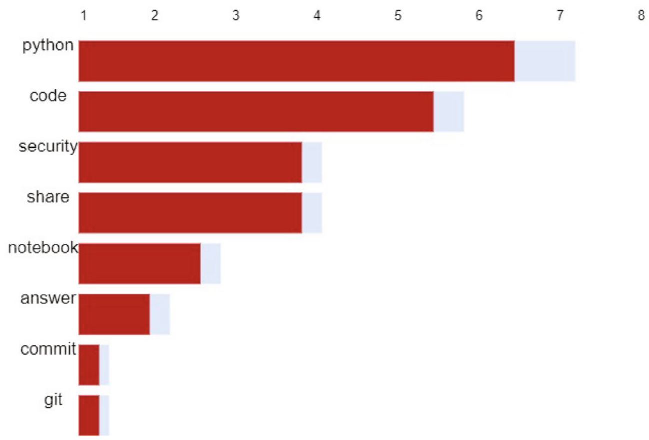


FIGURE 10 | Best topics extracted based on comments and answers of highly reputed users.

- **Association with BPT and Reputation**
  We further present the correlation coefficients (r value) between the users' BPT and reputation scores in Table 10. The correlation coefficient between openness and reputation is approximately 0.09, and the p value (<0.05) is statistically significant. Our study shows that users with high openness scores likely hold a high reputation in SO. We also computed a correlation between the users' openness trait and receiving up/down votes (which is the main metric of the users' reputation score) from their answers. Figure 11 shows the upvote to downvote ratio of SO users whose openness score falls above the third quartile among all users. Similarly, conscientiousness has a positive correlation (< 0.05), which indicates that users who have a high score in conscientiousness are likely to possess a high reputation score. On the contrary, neuroticism has a negative correlation with reputation, denoting that developers who have negative traits such as anger, sadness, agitation, and so forth, which may likely experience a negative impact on their reputation. Among the BPT, agreeable and extraversion traits do not show any significant correlation ($p > 0.05$) to the reputation score. Furthermore, we found no correlation between BPT and the users' patterns of asking questions.

- **Association With BPT and Textual Content of SO Answers/Comments:** We additionally investigated how the BPT may shape the topics of discussion in SO by analyzing their answers and comments using a fusion-based approach which is different from that applied in RQ3 (O1). We built the BERT-based embeddings [72] of the users' answers and comments and then reduced the dimensionality of the embeddings. Later, we created centroid-based clustering over the outcomes of the embeddings to find different groups of topics. For our experiments, we considered the number of topics to be 10, which provided the best outcomes. To find the best topics of discussion for the different BPTs, we kept the hyperparameters of LDA consistent. Users with a high openness BPT score are likely to enjoy intellectual and thought-provoking discussions. They prefer topics that include 'work', 'answer', 'question', 'solution', and 'observe'. This suggests their desire to explore, question established ideas, and seek new solutions. Conscientious individuals often participate in discussions involving practicality and organization. Topics such as 'use,' 'see,' 'need,' and 'example' are commonly found in their conversations, demonstrating their preference for clear instructions,

TABLE 10 | Association of the Big5 personality traits and the users' reputation.

Big5 personality traits	r (reputation)	p (reputation)
Openness	0.0992	**0.02346***
Conscient	0.0738	**0.0364***
Extravert	-0.0259	0.39901
Agreeable	0.0248	0.4143
Neuroticism	-0.0685	**0.0447***

Notes: Statistically are correlated (in bold). Significant correlations are shown in * where $p < 0.05$.

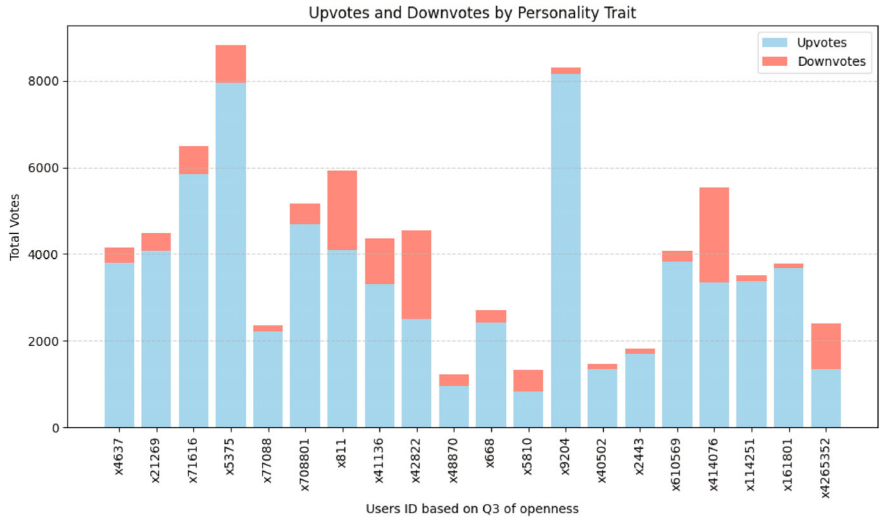


FIGURE 11 | Upvote to downvote ratio of posts from users who express more openness.

examples, and applicable solutions when they learn or solve any programming problem. Lastly, individuals with high extraversion BPT scores often participate in discussions or social interactions. Topics such as 'think', 'just', 'support', 'know', and 'want' are commonly found in their conversations, indicating their eagerness to share thoughts, support opinions, and engage with others in the community.

6 | Implications and Discussions

This section covers the main implications, the convergence of ML and psychology research, conclusions, and future directions for our study.

6.1 | Key Findings

We summarize our key findings as follows:

- Our results show that developers who have high openness scores are likely to be inclined toward mobile platforms, while developers who have high conscientious scores lean more toward OOP. While extraversion may be in line with collaborative web projects, the impact of agreeableness and neuroticism seems to be negligible.
- Our article explores how reputation is influenced by BPT. We discovered the importance of openness and conscientiousness traits in relation to community reputation ($p <$ 0.5, which is statistically significant). However, extraversion, agreeableness, and neuroticism traits do not exhibit a significant association, highlighting the complexity of assessing personality in this situation.
- BPTs represent a significant variance in the Twitter and SO-based datasets. Twitter represents the users' comprehensive behavior, that is, BPT, while SO only shows their behavior in a niche professional community. SO restricts the expression of emotions and personalities, whereas Twitter allows for a broader range of subjects and opinions, leading to distinct patterns in behavior.
- We also explored the relationship between the behavior of SO users and their reputation. SO answers may get upvoted or downvoted. Although upvotes may act as a reward, the likelihood of getting downvoted can deter people from answering. Developers who are more open-minded like to take on this as a challenge and reply to posted questions. Likewise, someone who is more conscientious pays more attention to the details, which can lead to a better in-depth understanding of the domain. As answering posted questions/problems requires mastery of their domain, conscientiousness can explain the higher correlations.

6.2 | Convergence of ML and Psychological Studies

The findings of our research have explored the complex landscape where BPT and users' behavior converge, particularly within the context of the software development community (i.e., question, answer, and reputation in SO) [24]. Ahmed et al. [108] described that in the collaborative environment of software development, soft skills like personality traits, social interaction skills, communication, and personal habits increase the likelihood of success of an individual and positively contribute to the project's overall aim. In a nutshell, our exploration of the intersection of psychology and technology is not just theoretical. It is a practical guide for companies aiming to build highly effective software development teams [109]. Organizations can position themselves for long-term success in a fast-moving market by identifying the hidden combination of personality traits, reputation, and domain-specific preferences [110, 111].

By using effective machine learning methods, our research is in line with the wider movement of interdisciplinary investigations at the interface between computer science and psychology. Our findings are comparable with general social structures, demonstrating the increasing acknowledgment that individual differences influence digital experiences and career results [112, 113]. This study adds new perspectives to the fields of software engineering and human-computer interaction (HCI) [114], while also complementing and expanding the scope of research in the fields of ML, DL, online behavior analysis, and personality psychology.

6.3 | Practical Implications

We propose a few practical implications regarding AI-based prediction models for finding the relationship between user BPT, online activity, and programming preferences.

- Employers and recruiters can build an automated tool that can facilitate the hiring of suitable candidates based on their project specifications by exploring the candidates' public tweets. However, if the candidate has no Twitter profile, the technique may still be applicable to their open-ended writings and word use patterns.
- Since we have demonstrated an association between BPT and programming preferences, different stakeholders (i.e., recruiters, policymakers, job agencies) can also conduct a manual survey (i.e., IPIP, TIPI) to recommend that users to be involved in a suitable programming platform if available tweets or open-ended writing content is not found.
- Technologies are upgrading on a continuous basis, and new technologies and paradigms are often introduced. In our article, we divided the technologies into four basic aspects: object construction, web-based platforms, technologies for handheld devices, and overall data storage and manipulating systems. However, based on our categorization, we can distribute the new technologies to their respective classes if such platforms are found.
- Since we discovered the association between BPT and community activities such as questioning, answering, and reputation, our method may also be able to play an important role in distributing tasks among the team members during project responsibility assignments.
- Since BPT is among the powerful instruments in psychology research, we can consider these traits to be a middle layer to map with another paradigm. For example, we may deploy gamification, metaverse, interactive simulation, and so forth, to understand the users' BPT, which can be mapped using programming preferences. In the literature, there are several studies [115] that predict the users' BPT using a gamification-based approach.

6.4 | Future Directions

In light of our findings and limitations, we can extend our research into several future avenues.

- *Content Type for Prediction*: In our article, we mainly predict the users' BPT from their writing pattern, that is, the text, used in Twitter using IBM personality insight API. However, in recent times, several studies have also used multi-modal content types such as image [116], EEG from brain signals [117], gait analysis from video [118], speech [119] and many more to predict the users' BPT. We may find there to be an association with the users' programming preferences when extracted from multi-modal content types for the better comprehension of user behavior.
- *Traits of Psychological Behavior*: In our study, we mainly found the users' programming platform recommendations to stem from their BPT. However, we may consider other important human behaviors such as values, self-efficacy, emotion, and sentiment when looking for an inclination with their programming platforms and technologies. In recent times, people are also bearing in mind how differently their brain works, as neurodiversity can be associated with BPTs and consequently, how they are correlated with the users' programming language recommendation.
- *Use of Cutting Edge Technologies*: Although we have demonstrated deep learning and the explainability of the features, we may still use further advanced deep learning-based techniques such as attention, large language models (LLM), and transformer-based approaches in the computational modules. Furthermore, we can use architectural features, that is, degree centrality, betweenness, clique relationship among the users, similarity among the users, and so forth, which may produce different relationships with the users' programming recommendations.
- *Dynamics of BPT*: We can analyze long-term data to evaluate if/how BPT and preferences change/evolve over time. In addition to long-term, short-term studies can also help identify trends.

7 | Conclusion

Our study has explored the intriguing relationship that exist between the users' BPT and programming languages within the framework of online communities. The various predictive models showed impressive accuracy, highlighting their potential for real-world programming language recommendation applications. Examining the BPT traits revealed how various traits affect the pattern of user involvement. We have conducted an analysis of platform differences between SO and Twitter, which revealed subtle aspects of user behavior. Furthermore, the complex interrelationship between asking, answering, and reputation on SO has demonstrated the dynamic nature of the knowledge-sharing ecosystem. Furthermore, a more thorough analysis of the complex relationships between BPTs and online behavioral dynamics could improve our understanding of user interactions.

Author Contributions

Md. Saddam Hossain Mukta: Conceptualization (lead); writing—original draft (lead); formal analysis (lead). **Badrun Nessa Antu**: Software (lead); review and editing. **Nasreen Azad:** Review and editing. **Iftekharul Abedeen:** Software and review. **Najmul Islam:** Conceptualization and supervision.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

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